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The Editors
Biometrical Journal

Dear Editors,

I am pleased to submit the manuscript “**Null-Model Choice for Rule-Defined Pattern Counts in Clustered Longitudinal Sequences: A Case Study of Jewish Halachic Menstrual Anticipation Rules**” for consideration as a **Case Study** in *Biometrical Journal*.

The problem. Jewish religious law prescribes that a woman anticipate her next menstruation from recurring arithmetic patterns in her cycle history (the *vesset* rules). Each rule defines an exact event — three equal consecutive intervals, a constant first or second difference, a repeated weekly anchor — and the natural question is whether such events occur in real cycle data more often than a suitable notion of chance predicts. The statistical difficulty is that the answer depends entirely on what “chance” is taken to mean: the data are 1,554 cycles from 118 women (5–45 cycles each) from the publicly archived Marquette University dataset (Fehring, 2012), collected in the randomised trial of Fehring et al. (*Contraception*, 2013); the pattern counts are integer-valued, structurally dependent by definition (one rule’s event set is contained in another’s), and both between-woman heterogeneity in regularity and within-woman ordering can generate them.

What the paper does. We compare three exchangeability nulls for the same five-count vector: a global permutation null (cycle lengths exchangeable across women), a multinomial i.i.d. null, and a within-woman permutation null that conditions on each woman’s observed multiset of cycle lengths and randomises only their order. All three are analysed with the same marginal Monte Carlo tests and a joint quadratic-form (Mahalanobis) permutation statistic whose covariance is estimated from the null replicates. The two global nulls reject strongly (joint $p < .001$); the within-woman null does not (joint $p = .13$), and all five observed counts fall at or below their within-woman null means. The paper shows why this contrast arises and what each null does and does not test. A compact simulation study reports the empirical size of every test used (including a split-batch check on the covariance estimate), the power of the within-woman tests against autoregressive and exact-repetition alternatives, and the sensitivity of the conclusions to unequal follow-up (truncation and leave-one-woman-out). An exploratory between-woman pairwise analysis is provided as Supporting Information.

Why a Case Study. The statistical tools are established — conditional randomisation, exchangeability blocks, quadratic-form permutation statistics — and the manuscript does not claim a new method. Its contribution is the careful construction and contrast of scientifically distinct conditional nulls for rule-defined pattern counts in clustered longitudinal sequences, the demonstration that the same data give diametrically different answers under population-level and within-subject exchangeability, and the validation of the procedures by simulation. The lesson transfers to other settings in which deterministic rules

are applied to recurrent-event sequences (for example, exact inter-seizure or inter-episode intervals), and this is the first formal randomisation-based test of the halachic rules on empirical data of which we are aware.

Required declarations.

Author concurrence: The sole author concurs with this submission.

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Prior submission and publication status: This manuscript has not been submitted elsewhere, is not under review at any other journal, and will not be published elsewhere within one year after its publication in *Biometrical Journal*.

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Ethics: This study is a secondary analysis of a publicly archived, anonymised dataset (Fehring, 2012; collected in the trial of Fehring et al., 2013). IRB approval was obtained for the original data collection by the original investigators. No new data were collected and there was no direct patient contact in this study.

Patient consent: No new data were collected and there was no contact with participants. According to the archive record of the dataset (Fehring, 2012), the human-subject data were anonymised for dissemination and data reuse was agreed to by the participants in the original consent form.

Conflicts of interest: The author declares no conflicts of interest.

Closely related work: A descriptive examination of vessel patterns appears in the author's PhD dissertation (Ariel University, 2022, in Hebrew), which also introduced the multinomial null. The within-woman permutation null, the joint quadratic-form test, the simulation validation, and the pairwise dependence analysis did not appear there. Related material on the halachic rules also appears in the author's accepted but not yet published article "Statistical Analysis for the Religious Laws of Abstinence before Menstruation" (*B.D.D.*; accepted 3 September 2021; in Hebrew), which addresses the abstinence rules and does not test pattern frequencies against a randomisation null; it is cited in the manuscript for the halachic haflaga convention. The dissertation and the accepted 2021 article (both in Hebrew) are submitted as supplementary material for review only.

Data and code availability. All analysis code, the archived source file, the filtered dataset and the script that derives it, and the cached Monte Carlo replicates needed to reproduce every number in the paper without re-simulation are publicly available at: <https://github.com/dvirross/VesselVectorTest>.

The original dataset is archived at Marquette University e-Publications (Fehring, 2012): https://epublications.marquette.edu/data_nfp/7/. A Reproducible Research ZIP will be supplied at the revision stage.

I believe this manuscript will be of interest to the readership of *Biometrical Journal*, and I look forward to the editorial team's consideration.

Yours sincerely,

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